

Chapter 3: Knowledge Representation and Probabilistic Reasoning

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Syllabus

- ❖ 3.1 Knowledge-based Agent
- ❖ 3.2. Knowledge Representation Techniques and Issues in Representation
- ❖ 3.3. Propositional & Predicate Logic
- ❖ 3.4. Semantic networks and frames
- ❖ 3.5. Probabilistic Reasoning, Inference using Bayes' Theorem
- ❖ 3.6. Fuzzy logic: membership functions, fuzzy inference systems

Marks Distributions: 14

Hours: 7 hrs

Knowledge

- ❖ Knowledge is “the sum of what is known: the body of truth, information, and principles acquired by mankind.” Or, "Knowledge is what I know, Information is what we know”.
- ❖ Knowledge is "information combined with experience, context, interpretation, and reflection. It is a high-value form of information that is ready to apply to decisions and actions." (T. Davenport et al., 1998)
- ❖ Knowledge consists of information that has been: interpreted, categorised, applied, experienced and revised.
- ❖ In general, knowledge is more than just data, it consist of: facts, ideas, beliefs, heuristics, associations, rules, abstractions, relationships, customs.

Types of Knowledge

❖ **Declarative Knowledge**

- Declarative knowledge is to know about something. It includes concepts, facts, and objects.

❖ **Procedural Knowledge**

- Procedural knowledge is a type of knowledge which is responsible for knowing how to do something.
- It includes rules, strategies, procedures, agendas, etc.

❖ **Heuristic Knowledge**

- Knowledge about the other types of knowledge is called Meta-knowledge.

❖ **Meta-Knowledge**

- Heuristic knowledge is representing knowledge of some experts in a field or subject

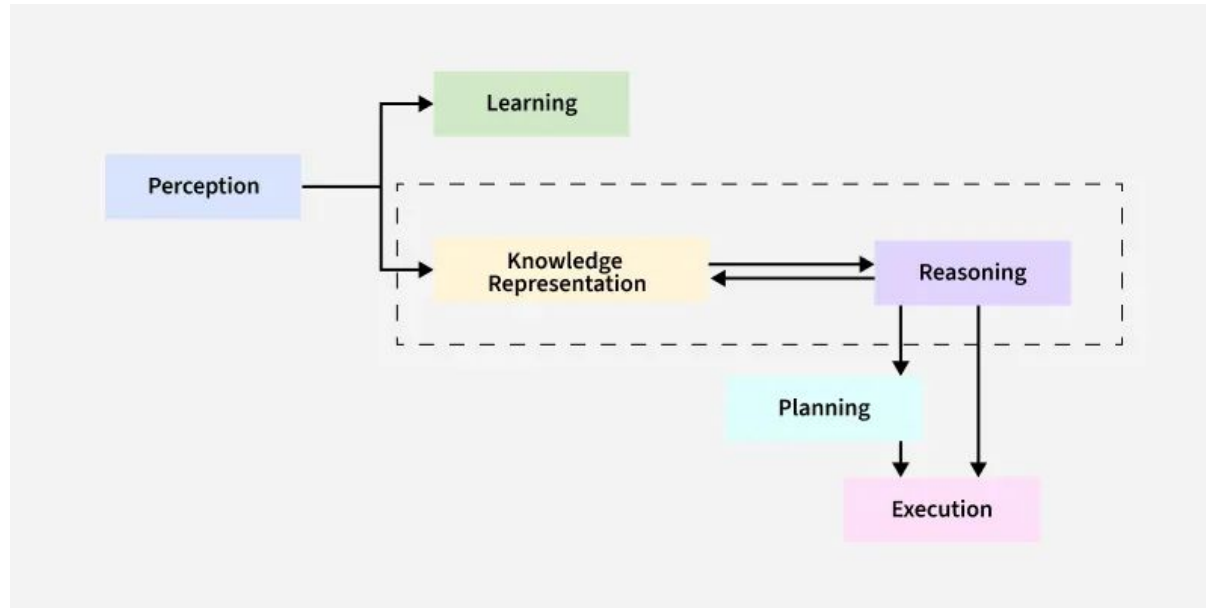
❖ **Structural Knowledge**

- Structural knowledge is basic knowledge to problem-solving.

Knowledge Representation and Reasoning

- ❖ The key factor to make machine intelligent is Knowledge
- ❖ Knowledge: facts, skills gained through education and experience (Human)
- ❖ Intelligence: ability to use knowledge
- ❖ Knowledge representation is concerned with encoding the human knowledge into the form that is efficiently manipulated by the computer.
- ❖ Knowledge Representation is the method used to encode knowledge in Intelligent Systems.
- ❖ A successful representation of some knowledge must, then, be in a form that is understandable by humans, and must cause the system using the knowledge to behave as if it knows it.

Knowledge Representation



Knowledge-based Agent

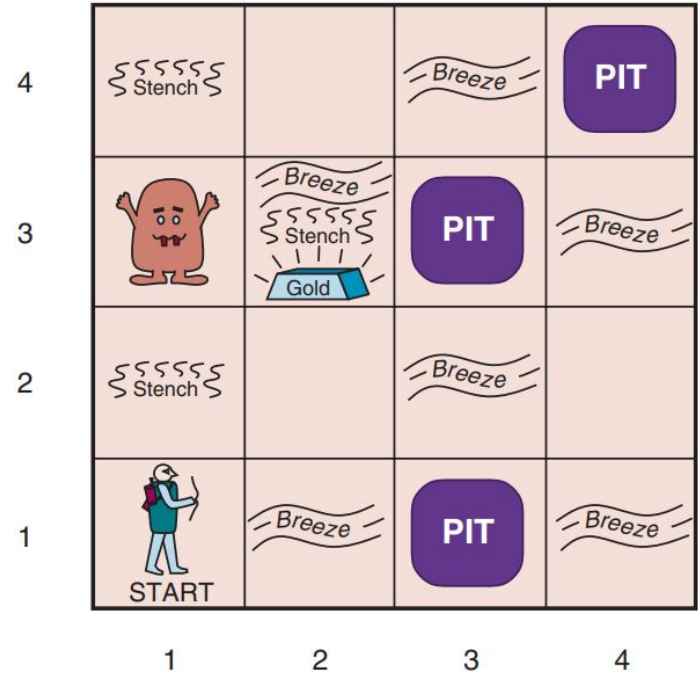
- ❖ A Knowledge-Based Agent is an intelligent system that maintains an internal state of knowledge, reasons over that knowledge, and acts based on that reasoning.
- ❖ Knowledge Base (KB): A set of sentences (facts and rules) expressed in a formal language (Logic).
- ❖ Inference Engine: The logical mechanism that derives new knowledge from the existing KB.
- ❖ How it Works
 - **TELL (Perceive):** The agent receives data from sensors and TELLS the KB the new facts (e.g., "I feel a breeze")
 - **ASK (Reason):** The agent ASKS the Inference Engine what action it should take based on its current knowledge.
 - **ACT:** The agent performs the action and TELLS the KB that it executed that action.

Knowledge-Based Agents: Working

- ❖ Each time the agent program is called, it does three things.
 - First, it TELLS the knowledge base what it perceives.
 - Second, it ASKS the knowledge base what action it should perform.
 - In the process of answering this query, extensive reasoning may be done about the current state of the world, about the outcomes of possible action sequences, and so on.
 - Third, the agent program TELLS the knowledge base which action was chosen, and returns the action so that it can be executed.

The Wumpus World Example

- ❖ **Grid:** A 4x4 grid of rooms. The agent starts at square [1,1].
- ❖ **The Hazards:**
 - Wumpus: A beast that eats the agent.
 - Pit: A bottomless hole the agent falls into.
- ❖ **The Goal:** Find the Gold and climb out safely.
- ❖ **Working:**
 - Agent is at [1,1] (safe)
 - Agent moves right to [2,1] (Breeze). There is a Pit in either [2,2] OR [3,1]
 - Go back to [1,1] (play it safe)
 - Agent moves up to [1,2] (stench)
 - Wumpus is in [1,3] OR [2,2]
 - The agent combines the knowledge from Step 1 and Step 2
 - Assumption: The game usually has one Wumpus and multiple Pits. If [2,2] had a Wumpus, there would be a Stench at [2,1], but there wasn't!
 - [2,2] does NOT have a Wumpus. Therefore, the Wumpus MUST be in [1,3].
 - Action: The agent can safely shoot its arrow toward [1,3] to kill the Wumpus.



Knowledge Engineering

- ❖ Just like "Software Engineering" is the process of building software, Knowledge Engineering (KE) is the process of building a Knowledge Base (KB).
- ❖ The Team:
 - Domain Expert: The human who knows the topic (e.g., a Doctor).
 - Knowledge Engineer: The AI expert who translates the Doctor's knowledge into code/logic.
- ❖ The 5-Step Process (The KE Lifecycle):
 - **Identify the Task:** What is the goal?
 - **Assemble Knowledge:** Interview the expert to gather facts.
 - **Decide a Vocabulary:** Choose the keywords/names. (e.g., Define Fatigue, Iron_Level, Hemoglobin).
 - **Encode the Knowledge:** Write the rules in logic.
 - **Debug/Refine:** Test the system. If it makes a mistake, fix the rules.

Knowledge Representation: Properties

- ❖ **Representational Adequacy** - The ability to represent the required knowledge;
- ❖ **Inferential Adequacy** - The ability to manipulate the knowledge represented to produce new knowledge corresponding to that inferred from the original;
- ❖ **Inferential Efficiency** - The ability to direct the inferential mechanisms into the most productive directions by storing appropriate guides;
- ❖ **Acquisitional Efficiency** - The ability to acquire new knowledge using automatic methods wherever possible rather than reliance on human intervention.

Techniques of Knowledge Representation

- ❖ Logical representation Scheme
- ❖ Procedural representation Scheme
- ❖ Network representation Scheme
- ❖ Structured representation Scheme

Logical Representation

- ❖ It uses expression in the form of formal notation to represent the knowledge base.
- ❖ Propositional logic and predicate logic are examples of the logical representation Scheme.
- ❖ Each sentence can be translated into logics using syntax and semantics.
- ❖ Syntax
 - Syntaxes are the rules which decide how we can construct legal sentences in the logic.
 - It determines which symbol we can use in knowledge representation.
 - How to write those symbols.

Logical Representation

- ❖ Semantics:
 - Semantics are the rules by which we can interpret the sentence in the logic.
 - Semantic also involves assigning a meaning to each sentence.
- ❖ Example
 - Propositional Logic: $P \rightarrow Q$ (If it rains, the ground will be wet.)

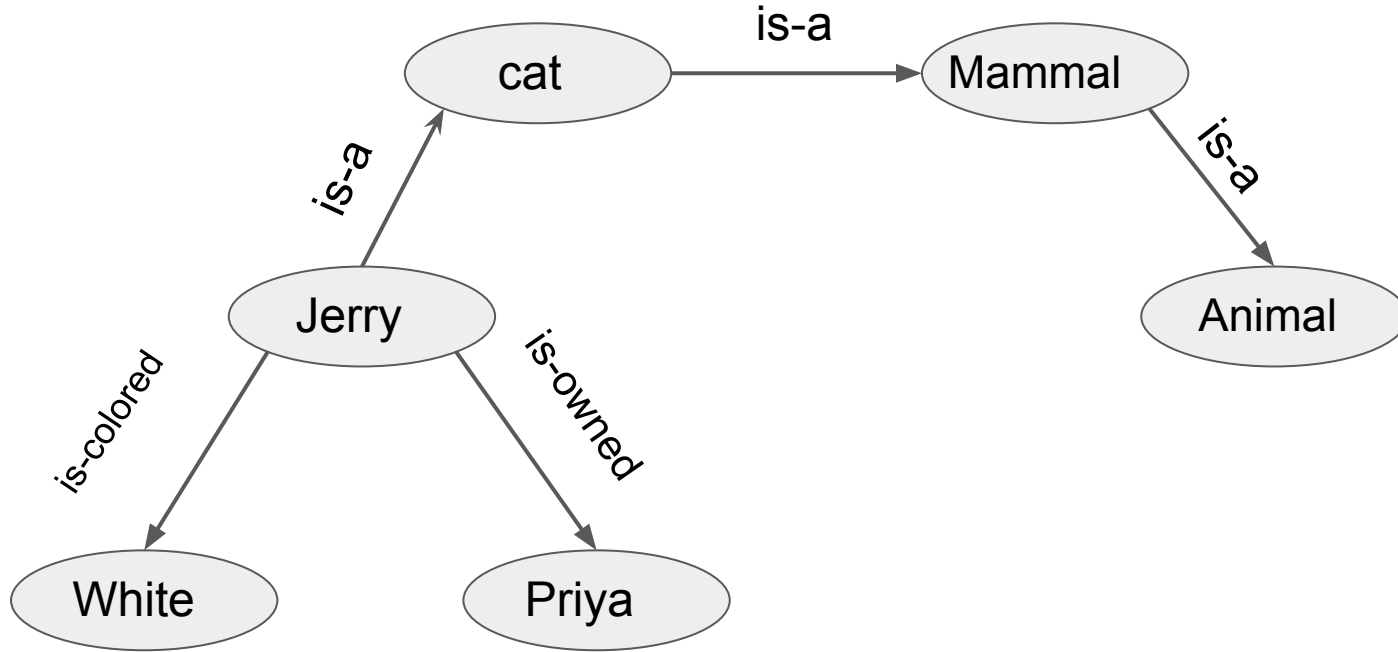
Procedural Representation

- ❖ Production Rules in Expert Systems
- ❖ Production rules system consist of (condition, action) pairs which mean, "If condition then action"
- ❖ Examples
 - IF (at bus stop AND bus arrives) THEN action (get into the bus)
 - IF (on the bus AND paid AND empty seat) THEN action (sit down).
 - IF type of light is sunlight THEN light is bright

Network Representation

- ❖ Network representation schemes represents knowledge in the form of a graph in which the nodes represent objects, situations, or events, and the arcs represent the relationships between them.
- ❖ Semantic networks, conceptual dependencies and conceptual graphs are examples of network representation.
- ❖ Statements
 - Jerry is a cat.
 - Jerry is a mammal
 - Jerry is owned by Priya.
 - Jerry is brown colored.
 - All Mammals are animal.

Network Representation



Structured Representation

- ❖ This is the extension of network representation schemes where nodes are complex data.
- ❖ Structures consisting of named slots with attached values.
- ❖ Scripts, frames and objects are examples of structured representation scheme.
- ❖ Frame for a Car
- ❖ Car:
 - Make: Toyota
 - Model: Corolla
 - Year: 2022
 - Owner: John Doe

Issues in Representation

- ❖ Important Attributes
 - Not all facts are equal. The system must filter out irrelevant data to save processing power.
 - If you choose too many attributes, the system becomes slow. If you choose too few, the system cannot make decisions.
- ❖ Relationships & Inverses
 - Facts are interconnected. The issue is choosing between Storage Space (Memory) and Processing Speed (CPU).
 - The engineer must decide if the system has more memory or more speed.
- ❖ Granularity
 - How detailed should the knowledge be?
 - If the granularity is too low, the system makes mistakes. If it's too high, the system gets slow and confused.
- ❖ Representation of Sets
 - How do we represent a group of things without writing down every single name?
 - We need a way to define "Classes" and "Categories" (Intension) rather than just listing "Members" (Extension)

Propositional Logic

- ❖ Propositional logic (PL) is the simplest form of logic where all the statements are made by propositions.
- ❖ A proposition is a declarative statement which is either true or false.
- ❖ It is a technique of knowledge representation in logical and mathematical form.

Propositional Logic

❖ Example

a) It is Sunday.

b) The Sun rises from West (False proposition)

c) $3+3=7$ (False proposition)

d) 5 is a prime number

e) "It is raining today, and street is wet."

f) "Ankit is a doctor, and his clinic is in Kathmandu."

Facts about Propositional Logic

- ❖ Propositional logic is also called **Boolean logic** as it works on 0 and 1.
- ❖ In propositional logic, we use symbolic variables to represent the logic, and we can use any symbol for a representing a proposition, such A, B, C, P, Q, R, etc.
- ❖ Propositions can be either **true** or **false**, but it cannot be both.
- ❖ Propositional logic consists of an **object**, **relations or function**, and **logical connectives**.
- ❖ These connectives are also called **logical operators**.
- ❖ The propositions and connectives are the basic elements of the propositional logic.
- ❖ A proposition formula which is always true is called **tautology**, and it is also called a **valid sentence**.
- ❖ A proposition formula which is always false is called **Contradiction**.
- ❖ Statements which are questions, commands, or opinions are not propositions such as "Where is Rohit", "How are you", "What is your name", are not propositions.

Proposition Logic

- ❖ Atomic Proposition
- ❖ Compound/Complex Proposition

Logical Connectives

Connective Symbols	Words	Terms	Example
$\wedge$	AND	Conjunction	$A \wedge B$
$\vee$	OR	Disjunction	$A \vee B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\Leftrightarrow$	If and only If	Biconditional	$A \Leftrightarrow B$
$\neg$	Not	Negation	$\neg A$

Logical Connectives

- ❖ **Negation:** A sentence such as $\neg P$ is called **negation** of **P**.
- ❖ **Conjunction:** A sentence which has $\wedge$ connective such as, $P \wedge Q$ is called a conjunction.
- ❖ **Example:** Rohan is intelligent **and** hardworking. It can be written as,
 - P = Rohan is intelligent,
 - Q = Rohan is hardworking.
 - Represented as $P \wedge Q$.

Logical Connectives

- ❖ **Disjunction:** A sentence which has $\vee$ connective, such as $P \vee Q$. is called **disjunction**, where P and Q are the propositions.

Example: "Rima is a doctor or Engineer"

Here $P =$ Rima is Doctor. $Q =$ Rima is Doctor,

so we can **write it as $P \vee Q$** .

Logical Connectives

- ❖ **Implication:** A sentence such as $P \rightarrow Q$, is called an **implication**. Implications are **also** known as **if-then rules**. It can be represented as

Example If it is raining, **then** the street is wet.

Let P = It is raining, and Q = Street is wet,

so it is **represented as** $P \rightarrow Q$.

- ❖ **Biconditional:** A sentence such as $P \Leftrightarrow Q$ is a Biconditional sentence,

Example I am breathing **if and only if** I am alive

P = I am breathing, Q = I am alive,

it can be represented as $P \Leftrightarrow Q$.

Order of Precedence of logical connectives

$\neg$
$\wedge$
$\vee$
$\rightarrow$
$\leftrightarrow$



High

Example:

$$\neg P \vee Q \wedge R \rightarrow S$$

equivalent to

$$((\neg P) \vee (Q \wedge R)) \rightarrow S$$

Low

Truth Table

- ❖ In propositional logic, we need to know the truth values of propositions in all possible scenarios.
- ❖ We can combine all the possible combination with logical connectives, and the representation of these combinations in a tabular format is called **Truth table**.

Truth Table

OR		
P	Q	$P \vee Q$
F	F	F
F	T	T
T	F	T
T	T	T

AND		
P	Q	$P \wedge Q$
F	F	F
F	T	F
T	F	F
T	T	T

Truth Table

if .. then .. ($\rightarrow$)		
P	Q	$P \rightarrow Q$
F	F	T
F	T	T
T	F	F
T	T	T

... if and only if .. ($\Leftrightarrow$)		
P	Q	$P \Leftrightarrow Q$
F	F	T
F	T	F
T	F	F
T	T	T

Logic: Propositional Logic: Example Sentences

- **P** means "It is hot."
- **Q** means "It is humid."
- **R** means "It is raining."
- **$P \wedge Q \rightarrow R$**
 - "If it is hot and humid, then it is raining."
- **$Q \rightarrow P$**
 - "If it is humid, then it is hot."
- **Q**
 - "It is humid."

Classwork:

1. $((\neg P) \vee (Q \wedge R))$

2. $(P \rightarrow Q) \wedge (Q \rightarrow R)$

Logic

- ❖ Logic is a **formal language** for **representing knowledge** such that **conclusions** can be drawn.
- ❖ **Logic** makes **statements** about **the world** which are **true (or false)**
- ❖ Compared to **natural languages** (expressive but context sensitive) and **programming languages** (good for concrete data structures but not expressive) logic combines the advantages of natural languages and formal languages.
- ❖ Logic is concise, unambiguous, expressive, context insensitive, effective for inferences.

Logic

- ❖ **Syntax:** Describe **possible configurations** that **constitute sentences**.
- ❖ **Syntax** is concerned with **the rules used** for **constructing**, or **transforming the symbols** and **words of a language**, as contrasted with the semantics of a language which is concerned with its meaning.
- ❖ **Semantics:** Determines what fact in the world, the sentence refers to i.e. the interpretation. Each sentence make claim about the world (meaning of sentence). **Semantic property include truth and falsity.**
- ❖ **Proof theory (Inference method):** Set of rules for generating new sentences that are necessarily true given that the old sentences are true.

Logic

- ❖ We will consider two kinds of logic: **propositional logic and first-order logic** or more precisely **first-order predicate calculus**.
- ❖ Propositional logic is of **limited expressiveness** but is useful to introduce many of the **concepts of logic's syntax, semantics and inference procedures**.

Syntax

- ❖ Syntax refers to the rules and structure governing the formation of valid expressions in a formal logic system.

- ❖ Example:

Valid: $P \wedge Q$

Invalid: $\wedge PQ$

Semantics

- ❖ **Semantics** deals with **the meanings assigned to symbols and formulas**.
- ❖ **In logic**, semantics defines **the truth or falsity** of propositions based on **interpretations or models**.
- ❖ Interpretation: P ="It rains", Q ="The ground is wet"
- ❖ Formula: $P \rightarrow Q$ means "If it rains, then the ground is wet."

Propositional Logic: Properties

- Validity (Tautology)
- Satisfiability (Contingency)
- Unsatisfiability (Contradictory)
- Entailment

Tautology

- ❖ A **formula** of **propositional logic** is a **tautology** if the formula itself is always **true regardless** of which valuation is used for the propositional variables.

Law of the Excluded Middle: $A \vee \neg A$

Law of Contraposition: $(A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A)$

Syllogism: $(A \rightarrow B) \wedge (B \rightarrow C) \rightarrow (A \rightarrow C)$

- ❖ There are also propositions that are always false is called **Contradiction**.
- ❖ A proposition that is neither a tautology nor a contradiction is called a **contingency**.

Logical Equivalence

Basic Logical Laws - Equivalences		
Null Laws	$P \wedge 0 \Leftrightarrow 0$	$P \vee 1 \Leftrightarrow 1$
Idempotent Laws	$P \vee P \Leftrightarrow P$	$P \wedge P \Leftrightarrow P$
Involution Laws	$\neg(\neg P) \Leftrightarrow P$	$\neg(\neg P) \Leftrightarrow P$
Identity Laws	$P \vee 0 \Leftrightarrow P$	$P \wedge 1 \Leftrightarrow P$
Negation Laws	$P \wedge \neg P \Leftrightarrow 0$	$P \vee \neg P \Leftrightarrow 1$
Absorption Laws	$P \wedge (P \vee Q) \Leftrightarrow P$	$P \vee (P \wedge Q) \Leftrightarrow P$
De Morgan's Laws	$\neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q)$	$\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$
Commutative Laws	$P \vee Q \Leftrightarrow Q \vee P$	$P \wedge Q \Leftrightarrow Q \wedge P$
Associative Laws	$(P \vee Q) \vee R \Leftrightarrow P \vee (Q \vee R)$	$(P \wedge Q) \wedge R \Leftrightarrow P \wedge (Q \wedge R)$
Distributive Laws	$P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$	$P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$

Validity

- ❖ A logically valid argument is one where the conclusion follows from the premises.
- ❖ An invalid argument is where the conclusion does not follow from the premises.
- ❖ A formula of a formal language is a valid formula if and only if it is true under every possible interpretation of the language.
- ❖ A less precise but intuitively clear way of putting this is to say that in a valid argument IF the premises are true, then the conclusion must be true.

Validity

An example of a valid argument is given by the following well-known **syllogism**:

Example: All men are mortal.
 Socrates is a man.
 Therefore, Socrates is mortal.

Validity

The following argument is of the same logical form but with false premises and a false conclusion, and it is equally valid:

Example: All women are cats.
 All cats are men.
 Therefore, all women are men.

Well Formed Formula (WFF)

- ❖ A Well-Formed Formula (WFF) is a syntactically correct expression in formal logic.
- ❖ It follows specific rules and conventions, ensuring it is meaningful within the logical system.
- ❖ The concept is fundamental in propositional and predicate logic, as only WFFs can be interpreted or used in reasoning.

Well Formed Formula (WFF)

- ❖ An atomic proposition (e.g., P) is a WFF, negation of P is also WFF.
- ❖ If A and B are WFFs, then conjunction, disjunction, implies, iff also WFFs.
- ❖ Parentheses are necessary to avoid ambiguity, except when the precedence of operators is clear.

Example

P (Atomic proposition, valid WFF).

$\neg P$ (Negation of P , valid WFF).

$(P \wedge Q)$ (Conjunction, valid WFF).

$((P \vee Q) \rightarrow R)$ (Implication with disjunction, valid WFF).

Well Formed Formula (WFF)

Examples(Not WFFs)

$P \wedge \vee Q$ (Invalid syntax: improper use of connectives).

$\neg \wedge P$ (Invalid syntax: missing a proposition after $\neg$).

PQR (Invalid syntax: no logical connectives or structure).

$(P \wedge Q \vee R$ (Invalid syntax: unbalanced parentheses).

Limitation of propositional Logic

- ❖ Can only represent knowledge as complete sentence.
- ❖ Can't analyse internal structure of sentence.
- ❖ Absence of quantifiers.
- ❖ Eg: All kings are men , All men are mortal.

Predicates

- ❖ A predicate is a symbol or function that represents a property or relationship that can be attributed to objects or individuals within domain.
- ❖ It can be thought of as a function that returns a truth value (true or false) depending on the input(s) it is applied to.
- ❖ Predicates are used in predicate logic to express statements about objects and their relationships.
- ❖ A predicate is a function that applies a property or relation to objects in a domain, and returns a truth value depending on the arguments.

Predicates

❖ Example

$P(x)$: "x is a prime number."

- $P(3)$ is true because 3 is a prime number.
- $P(4)$ is false because 4 is not a prime number.
- $S(A,B)$ means "A is the parent of B."
- $L(A,B,C)$ means "A loves B more than C."

Quantifiers

- ❖ Predicate calculus can use quantifiers for statement which express the expression as some object or all objects within given domain.
- ❖ Universal ($\forall$) and existential ($\exists$) quantifiers are used for expressing statement.
- ❖ In logic, quantifiers are symbols used to express the extent to which a predicate applies to elements in a domain.
- ❖ They are used in predicate logic (and first-order logic, or FOPL) to define the scope of variables in logical statements, indicating whether the statement applies to all elements or to at least one element in the domain.

Universal Quantifier ($\forall$)

- ❖ The universal quantifier $\forall$ is used to state that a predicate applies to every element in the domain of discourse.
- ❖ It is often read as "for all" or "for every."
- ❖ $\forall x P(x)$, which means "For all x , $P(x)$ is true."

Universal Quantifier ($\forall$)

- ❖ If $P(x)$ means "x is a human," then $\forall xP(x)$ means "All x's are humans."
- ❖ If the domain is the set of all people, the statement would assert that every person is a human
- ❖ $\forall x(\text{Human}(x) \rightarrow \text{Mortal}(x))$
- ❖ This reads: "For all x, if x is a human, then x is mortal."

Existential Quantifier ($\exists$)

- ❖ The existential quantifier $\exists$ is used to state that there exists at least one element in the domain for which the predicate is true.
- ❖ It is often read as "there exists" or "there is at least one."
- ❖ $\exists xP(x)$, which means "There exists an x such that $P(x)$ is true."
- ❖ If $P(x)$ means " x is a human," then $\exists x P(x)$ means "There exists at least one x who is a human."
- ❖ In a domain of people, this statement says that there is at least one human.
- ❖ $\exists x(\text{Human}(x) \wedge \text{Tall}(x))$
- ❖ This reads: "There exists an x such that x is a human and x is tall."

Predicate Logic

- ❖ Predicate logic (also known as first-order logic, first-order predicate logic or FOPL) is an extension of propositional logic that allows for more expressive and detailed statements about the properties of objects and the relationships between them.
- ❖ In predicate logic, we use predicates, quantifiers, and variables to express logical relationships.

First-order predicate logic or FOPL

- ❖ First-order logic is another way of knowledge representation in artificial intelligence. It is an extension to propositional logic.
- ❖ FOPL is sufficiently expressive to represent the natural language statements in a concise way.
- ❖ First-order logic is also known as Predicate logic or First-order predicate logic.
- ❖ First-order logic is a powerful language that develops information about the objects in a more easy way and can also express the relationship between those objects.
- ❖ A predicate is a function or relation that represents a property of objects or a relationship between objects.
- ❖ For example, $P(x)$ might mean "x is a human."
- ❖ Variables represent individual objects or elements in the domain.
- ❖ For example, In $P(x)$, x is a variable that can represent any individual.

First-order predicate logic or FOPL

- ❖ Quantifiers specify the extent or scope of a statement over the objects in the domain.
- ❖ A constant refers to a specific object in the domain.
- ❖ For example, a might refer to a specific individual, like "John."

Constant	1, 2, A, John, Mumbai, cat,....
Variables	x, y, z, a, b,....
Predicates	Brother, Father, >,....
Function	sqrt, LeftLegOf,
Connectives	$\wedge$, $\vee$, $\neg$, $\Rightarrow$, $\Leftrightarrow$
Equality	$=$
Quantifier	$\forall$, $\exists$

Interpretation

- ❖ A model or interpretation in predicate logic assigns meanings to the predicates, functions, and constants in a logical system.
- ❖ The interpretation maps the predicates to specific relations in the real world and the constants to specific objects.
- ❖ Let $\text{Human}(x)$ mean "x is a human."
- ❖ Let $\text{Mortal}(x)$ mean "x is mortal."

Advantages of FOPL

- Expressiveness
- Generalization
- Quantifications

Examples

- All birds fly. Predicate is "fly(bird)."

$\forall x \text{ bird}(x) \rightarrow \text{fly}(x).$

- Every man respects his parent. Predicate "respect(x, y)," where x=man, and y= parent.

$\forall x \text{ man}(x) \rightarrow \text{respects}(x, \text{parent}).$

- Some boys play cricket. Predicate Play(x, y)," where x= boys, and y= game

$\exists x \text{ boys}(x) \wedge \text{play}(x, \text{cricket}).$

Examples

- Not all students like both Mathematics and Science.
- There exists a person who is taller than 7 feet.
- For every human, there exists a country where they were born.

Examples

- Not all students like both Mathematics and Science. Predicate "like(x, y)," where x= student, and y= subject.

$\neg \forall (x) [\text{student}(x) \rightarrow \text{like}(x, \text{Mathematics}) \wedge \text{like}(x, \text{Science})]$.

- There exists a person who is taller than 7 feet.

$\exists x(\text{Person}(x) \wedge \text{TallerThan}(x, 7))$

- For every human, there exists a country where they were born.

$\forall x(\text{Human}(x) \rightarrow \exists y(\text{Country}(y) \wedge \text{BornIn}(x, y))$

Examples

- There is no one who is both a human and a non-mortal.
- For every student, there is a professor who teaches them.
- There is a person who is both a teacher and a student

Examples

- There is no one who is both a human and a non-mortal.

$$\neg \exists x(\text{Human}(x) \wedge \neg \text{Mortal}(x))$$

- For every student, there is a professor who teaches them.

$$\forall x(\text{Student}(x) \rightarrow \exists y(\text{Professor}(y) \wedge \text{Teaches}(y,x)))$$

- There is a person who is both a teacher and a student

$$\exists x(\text{Teacher}(x) \wedge \text{Student}(x))$$

Horn Clause

- ❖ A Horn clause is a disjunction of literals (which are atomic propositions or their negations) where at most one of the literals is positive (i.e., not negated).
- ❖ In other words, a Horn clause contains:
 - At most one positive literal (an atom).
 - Any number of negative literals (negations of atoms).

Example of horn Clause

- $(Q \wedge R) \rightarrow P$
- This can be written as a disjunction in a more formal way: $\neg Q \vee \neg R \vee P$
- This means: "If Q and R are both true, then P must be true."

Rules of inference

- Rules of inference are logical rules that dictate the valid steps to derive new information or conclusions from known facts or premises.
- These rules are fundamental to reasoning systems such as expert systems, automated theorem provers, and many forms of symbolic AI.
- The goal is to model human-like reasoning processes in machines.

Rule of Inferences

Modus Ponens

$$\frac{p \quad p \rightarrow q}{q}$$

Addition

$$\frac{p}{p \vee q}$$

Simplification

$$\frac{p \wedge q}{p}$$

Modus Tollens

$$\frac{\neg q \quad p \rightarrow q}{\neg p}$$

Resolution

$$\frac{p \vee q \quad \neg p \vee r}{q \vee r}$$

Conjunction

$$\frac{p \quad q}{p \wedge q}$$

Hypothetical Syllogism

$$\frac{p \rightarrow q \quad q \rightarrow r}{p \rightarrow r}$$

Disjunctive Syllogism

$$\frac{p \vee q \quad \neg p}{q}$$

Example

- Let p be “It is raining,”
- q be “I will make tea,”
- r be “I will read a book.”

Example: Modus Ponens

“If it is raining, then I will make tea.”

“It is raining.”

“Therefore, I will make tea.”

$p \rightarrow q$

p

$\therefore q$

Modus Ponens (Law of Detachment)

$((p \rightarrow q) \wedge p) \Rightarrow q$

Example — Modus Tollens

“If it is raining, then I will make tea.”

“I do not make tea.”

“Therefore, it is not raining.”

$$p \rightarrow q$$

$$\neg q$$

$$\therefore \neg p$$

Modus Tollens

$$\left((p \rightarrow q) \wedge \neg q \right) \Rightarrow \neg p$$

Example — Hypothetical Syllogism

“If it is raining, then I will make tea.”

“If i make tea, then I will read a book.”

“Therefore, if it rains, then I will read a book.”

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

Hypothetical Syllogism

$$\left((p \rightarrow q) \wedge (q \rightarrow r) \right) \Rightarrow (p \rightarrow r)$$

Example — Disjunctive Syllogism

“I will make tea, or I will read a book.”

$q \vee r$

“I will not make tea.”

$\neg q$

“Therefore, I will read a book.”

$\therefore r$

Disjunctive Syllogism

$\left((q \vee r) \wedge \neg q \right) \Rightarrow r$

Example — Addition

“I will make tea.”

q

Addition

“Therefore, I will make tea, or I will read a book.”

$\therefore q \vee r$

$q \Rightarrow q \vee r$

Example — Simplification

“I will make tea and I will read a book.”

$$q \wedge r$$

“Therefore, if it rains, then i will read a book.”

$$\therefore q$$

Simplification

$$(p \wedge q) \Rightarrow p$$

Example — Resolution

“It is raining, or I will make tea.”

$p \vee q$

“It is not raining, or I will read a book.”

$\neg p \vee r$

“Therefore, I will make tea, or I will read a book.” $\therefore q \vee r$

Resolution

$$\left((p \vee q) \wedge (\neg p \vee r) \right) \Rightarrow (q \vee r)$$

Example 2: For the set of premises "If I play hockey, then I am sore the next day." "I use the whirlpool if I am sore." "I did not use the whirlpool". What relevant conclusion can be drawn?

Explain the rules of inference used to draw the conclusion.

Solution:

Let p = "I play hockey", q = "I am sore", r = "I use the whirlpool"

Then the above premises are

a) $p \rightarrow q$

b) $q \rightarrow r$

c) $\neg r$

Using hypothetical syllogism in premises a and b we have $p \rightarrow r$ i.e. "if I play hockey, then use whirlpool"

Using the modus tollens in premise c and inferred proposition $p \rightarrow r$ we conclude p is true i.e. p is false. p is false means "I did not play hockey"

Example 1: Construct an argument using rules of inference to show that the hypotheses:

"If it does not rain or if it is not foggy, then the sailing race will be held and the life saving demonstration will go on," " If the sailing race is held, then the trophy will be awarded," and "The trophy was not awarded" imply the conclusion " It rained".

Solution:

Let p = "It rains", q = "It is foggy", r = "the sailing race is held", s = "Life saving demonstration is done", and t = "Trophy is awarded".

Then we have to show that the argument

$[((\neg p \vee \neg q) \rightarrow (r \wedge s)) \wedge (r \rightarrow t) \wedge \neg t] \rightarrow p$ is valid.

[1] $(r \rightarrow t)$ [Hypothesis]
[2] $\neg t$ [Hypothesis] $\longrightarrow (r \rightarrow t) \wedge \neg t$

[3] $\neg r$ [Modus Tollens using steps 1 and 2]

[4] $((\neg p \vee \neg q) \rightarrow (r \wedge s))$ [Hypothesis]

[5] $\neg(\neg p \vee \neg q) \vee (r \wedge s)$ [Implication of Step 4]

[6] $(p \wedge q) \vee (r \wedge s)$ [De Morgan's Law in Step 5]

[7] $p \vee (r \wedge s)$ [Simplification using step 6]

[8] $p \vee r$ [Simplification using step 7]

Here our original premises changes to $(p \vee r) \wedge \neg r$ [from step 8 and 3]

[9] $r \vee p$ [Commutative law in step 8]

[10] $\neg r$ [From step 3]

[11] p [Disjunctive syllogism using steps 9 and 10]

Hence argument is valid. With conclusion "It rained"

Disadvantages of Rule of Inferences

- As the number of rules and premises increases, applying them can become complex and error-prone.
- The number of possible inferences can grow exponentially, leading to computational inefficiency

Resolution (Resolution Refutation System (RRS))

- Resolution is a theorem proving technique that proceeds by building refutation (the action of proving a statement or theory to be wrong or false.) proofs, i.e., proofs by contradictions.
- It was invented by a **Mathematician John Alan Robinson** in the year **1965**.
- Resolution is used, if there are various statements are given, and we need to prove a conclusion of those statements.
- Unification (making expressions looks similar) is a key concept in proofs by resolutions.
- Resolution is a single inference rule which can efficiently operate on the conjunctive normal form or clausal form.

Steps for Resolution

- Conversion of facts into first-order logic (FOL).
- Convert FOL statements into CNF
- Negate the statement which needs to prove (proof by contradiction)
- Draw resolution graph (unification).

What is Clause ?

- Disjunction of literals (an atomic sentence) is called a clause. It is also known as a unit clause.
- A literal is an atomic proposition or its negation.
- Example $\neg P \vee Q$, $P \vee \neg R$
- In the first clause, " $\neg P$ " and " Q " are literals.

Two Normal (Canonical) Forms

All wffs can be expressed in the following two normal forms

1. CNF (Conjunctive Normal Form)

e.g.: $(A \vee \neg B) \wedge (B \vee \neg D \vee \neg C)$

clause 1 clause 2

2. DNF (Disjunctive Normal Form)

e.g.: $(A \wedge \neg B) \vee (B \wedge \neg D \wedge \neg C)$

models models

Conversion to CNF

- A sentence that is expressed as a conjunction of disjunctions of literals is said to be in conjunctive normal form (CNF).
- A sentence in CNF that contains only k literals per clause is said to be in k -CNF.

Algorithm

Eliminate $\leftrightarrow$ rewriting $P \leftrightarrow Q$ as $(P \rightarrow Q) \wedge (Q \rightarrow P)$

Eliminate $\rightarrow$ rewriting $P \rightarrow Q$ as $\neg P \vee Q$

Use De Morgan's laws to push $\neg$ inwards:

- rewrite $\neg (P \wedge Q)$ as $\neg P \vee \neg Q$
- rewrite $\neg (P \vee Q)$ as $\neg P \wedge \neg Q$

Eliminate double negations: rewrite $\neg \neg P$ as P

Use the distributive laws to get CNF:

- rewrite $(P \wedge Q) \vee R$ as $(P \vee R) \wedge (Q \vee R)$

Flatten nested clauses:

- $(P \wedge Q) \wedge R$ as $P \wedge Q \wedge R$
- $(P \vee Q) \vee R$ as $P \vee Q \vee R$

Example

$$B \Leftrightarrow (A \vee C)$$

- Eliminate $\Leftrightarrow$, replacing $\alpha \Leftrightarrow \beta$ with $(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$.
 - $(B \Rightarrow (A \vee C)) \wedge ((A \vee C) \Rightarrow B)$
- Eliminate $\Rightarrow$, replacing $\alpha \Rightarrow \beta$ with $\neg \alpha \vee \beta$.
 - $(\neg B \vee A \vee C) \wedge (\neg(A \vee C) \vee B)$
- Move $\neg$ inwards using de Morgan's rules and double-negation:
 - $(\neg B \vee A \vee C) \wedge ((\neg A \wedge \neg C) \vee B)$
- Apply distributivity law ($\wedge$ over $\vee$) and flatten:
 - $(\neg B \vee A \vee C) \wedge (\neg A \vee B) \wedge (\neg C \vee B)$

Exercise

- $\neg(P \wedge Q) \vee R$
- $\neg(P \vee Q) \wedge (R \rightarrow S)$

Resolution: Example

We first put each hypothesis in CNF:

$$1. \quad r \rightarrow u = (\neg r \vee u)$$

$$2. \quad u \rightarrow \neg w = (\neg u \vee \neg w)$$

$$3. \quad \neg r \rightarrow \neg w = (r \vee \neg w)$$

Resolution: Example

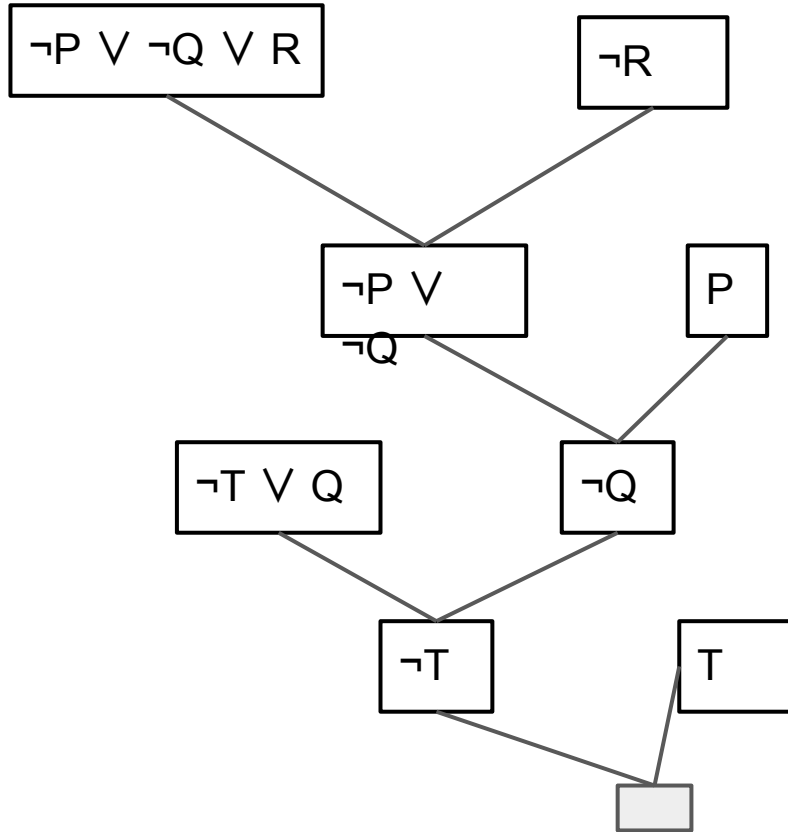
Proof (about Joe and his umbrella) using proof by contradiction with resolution:

1. $\neg r \vee u$ Premise
2. $\neg u \vee \neg w$ Premise
3. $r \vee \neg w$ Premise
4. w Negation of conclusion
5. $\neg r \vee \neg w$ Using resolution on 1, 2
6. $\neg w \vee \neg w$ Using resolution on 3, 5
7. $\neg w$ Using resolution on 4, 6
8. FALSE Using resolution on 4, 7

Resolution: Example

S. No.	Given Axioms	Convert to Clause Form
1.	P	P
2.	$(P \wedge Q) \rightarrow R$	$\neg P \vee \neg Q \vee R$
3.	$S \vee T \rightarrow Q$	$\neg S \wedge \neg T \vee Q$
4.	$T \rightarrow Q$	$\neg T \vee Q$
5.	T	T
Conclusion: R		

Resolution: Example



Universal instantiation (UI)

- Universal Instantiation (UI) is one of the most important rules in Predicate Logic (First-Order Logic).
- It is the rule that allows us to apply a **"General Law"** to a **"Specific Person/Object"**.
- If a statement is true for everything in the universe (indicated by the Universal Quantifier $\forall$)
- Example:

"All men are mortal."

$$\forall x (\text{Man}(x) \Rightarrow \text{Mortal}(x))$$

Existential Instantiation (EI)

- Existential Instantiation (EI) is the rule used to handle "Specific Unknowns."
- It allows us to eliminate the Existential Quantifier ($\exists$) so we can do logical reasoning without worrying about variables.
- Knowledge Base:

"There is a Crown on the table.

$$\exists x (\text{Crown}(x) \wedge \text{OnTable}(x))$$

Applying Existential Instantiation (EI):

Result:

$$\text{Crown}\{C_1\} \wedge \text{OnTable}(C_1)$$

Inference Engine: Forward and Backward Chaining

❖ Forward Chaining

- It begins with known facts (positive literals) in the Knowledge Base.
- It checks the implications (rules). If all premises of a rule are known, the conclusion is added to the facts.
- This process repeats until the query (q) is found or no new facts can be generated.

❖ Backward Chaining

- It works backward from the query (q).
- It finds implications where q is the conclusion.
- It then tries to prove the premises of those implications. This creates a chain of sub-goals until it reaches known facts.
- It is described as essentially identical to AND-OR Graph Search.

Example

Facts:

- A is true
- B is true

The Rules (Clauses):

1. $P \Rightarrow Q$
2. $L \sqcap M \Rightarrow P$
3. $B \sqcap L \Rightarrow M$
4. $A \sqcap P \Rightarrow L$
5. $A \sqcap B \Rightarrow L$

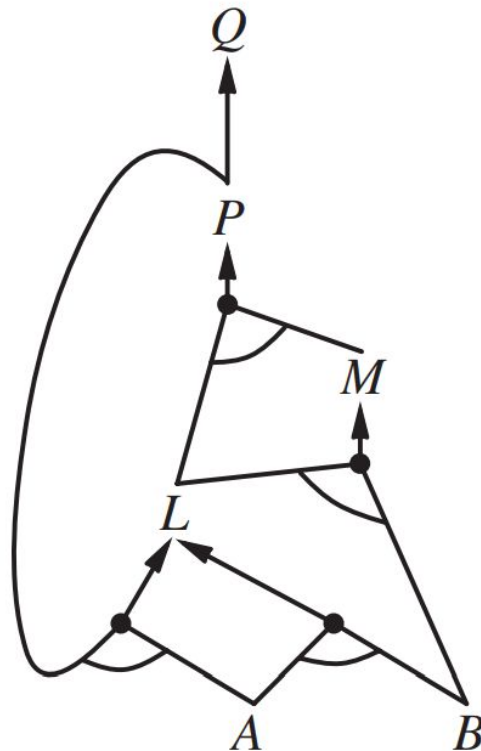
The Goal:

- Prove Q

Example:

Forward Chaining

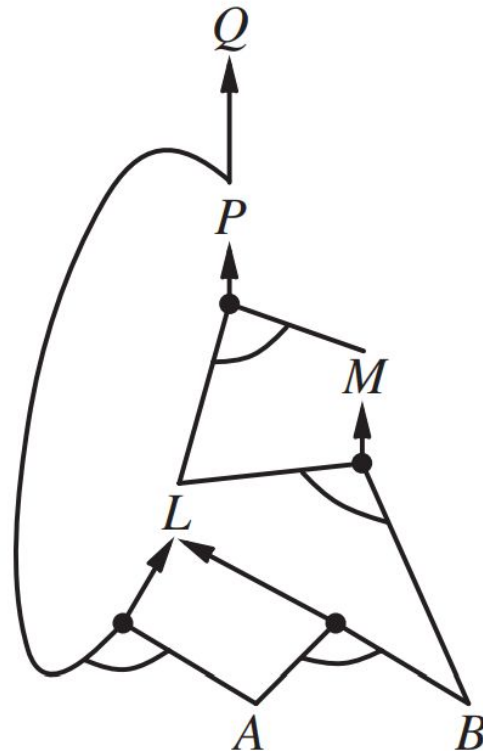
1. Initial State: Known: {A, B}, Target: Q
2. Iteration 1:
 - a. The algorithm sees Rule 5: $A \sqcap B \Rightarrow L$.
 - b. Since we have both A and B, the rule fires.
 - c. New Fact: L is added.
 - d. Known: {A, B, L}
3. Iteration 2:
 - a. Now we have L and B.
 - b. The algorithm sees Rule 3: $B \sqcap L \Rightarrow M$.
 - c. The rule fires.
 - d. New Fact: M is added.
 - e. Known: {A, B, L, M}
4. Iteration 3:
 - a. Now we have L and M.
 - b. The algorithm sees Rule 2: $L \sqcap M \Rightarrow P$.
 - c. New Fact: P is added.
 - d. Known: {A, B, L, M, P}
5. Iteration 4:
 - a. Now we have P. The algorithm sees Rule 1: $P \Rightarrow Q$
 - b. New Fact: Q is added.
6. Success! Q is found in the knowledge base.



Example:

Backward Chaining

1. Goal: Is Q true?
 - a. Check Graph:
 - b. To get Q, I need P. Sub-Goal: Prove P.
2. Goal: Is P true?
 - a. Check Graph: To get P, I need L AND M.
 - b. Sub-Goal: Prove L.
 - c. Sub-Goal: Prove M.
3. Branch 1 (Prove L):
 - a. Check Graph: To get L, I need A AND B.
 - b. Check Facts: Do I have A? Yes. Do I have B? Yes. Result:
 - c. L is verified.
4. Branch 2 (Prove M):
 - a. Check Graph: To get M, I need B AND L.
 - b. Check Facts: Do I have B? Yes. Do I have L? Yes (just verified in Branch 1).
 - c. Result: M is verified.
5. Conclusion:
 - a. Since L and M are true $\rightarrow$ P is true.
 - b. Since P is true $\rightarrow$ Q is true.



Review of Probability (The Basics)

- ❖ **Prior Probability ($P(A)$):** The probability of an event before we have any evidence.

Example: $P(\text{Flu}) = 0.1$ (10% of people have the flu).

- ❖ **Conditional Probability ($P(A|B)$):** The probability of A given that B has happened.

Example: $P(\text{Cough} | \text{Flu}) = 0.8$ (If you have the flu, there is an 80% chance you cough).

- ❖ **Joint Probability ($P(A \cap B)$):** The probability that both happen together.

Formula: $P(A \cap B) = P(A|B) * P(B)$.

- ❖ **Marginalization (Sum Rule):** The process of adding up probabilities to find the total chance of an event.

Bayes' Theorem

Bayes' Theorem is a rule for updating belief based on new evidence. It allows us to calculate the probability of a Cause given an Effect, using knowledge we already have about the Effect given the Cause.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \text{ where } P(B) \neq 0$$

Where:

- A (Hypothesis): What we want to prove (e.g., "Patient has Cancer").
- B(Evidence): The new data we observed (e.g., "Positive Lab Test").
- P(A) (Prior): Initial belief probability.
- P(B|A) (Likelihood): The probability of seeing this evidence if the hypothesis is true.
- P(B) (Marginal Evidence): The total probability of the evidence occurring (normalizing constant).

Example:

In County, 51% of the adults are males and the other 49% are females. One adult is randomly selected for a survey involving credit card usage. It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration).

- a) Find the prior probability that the selected person is a male.
- b) Use this additional information to find the probability that the selected subject is a male.

Solution:

Given Data:

Let M be the event "Subject is Male".

Let F be the event "Subject is Female".

Let C be the event "Subject smokes a Cigar"

From the question,

we have:

$P(M) = 0.51$ (51% are Males)

$P(F) = 0.49$ (49% are Females)

$P(C|M) = 0.095$ (9.5% of Males smoke cigars)

$P(C|F) = 0.017$ (1.7% of Females smoke cigars)

a) Find the prior probability that the selected person is a male.

Ans: The Prior Probability is simply the probability of the event before any specific evidence (like smoking) is considered. This is given directly in the problem statement.

$$P(M) = 0.51$$

b. Use this additional information to find the probability that the selected subject is a male.

Step 1: Calculate the Total Probability of Smoking ($P(C)$) using the Sum Rule (Marginalization):

$$P(C) = P(C|M) \cdot P(M) + P(C|F) \cdot P(F)$$

$$P(C) = (0.095 \cdot 0.51) + (0.017 \cdot 0.49)$$

$$P(C) = 0.04845 + 0.00833$$

$$P(C) = 0.05678$$

Step 2: Apply Bayes' Theorem

The probability that the selected
subject is male is
approximately 0.8533 (or 85.33%).

$$P(M|C) = \frac{P(C|M) \cdot P(M)}{P(C)}$$

$$P(M|C) = \frac{0.095 \times 0.51}{0.05678}$$

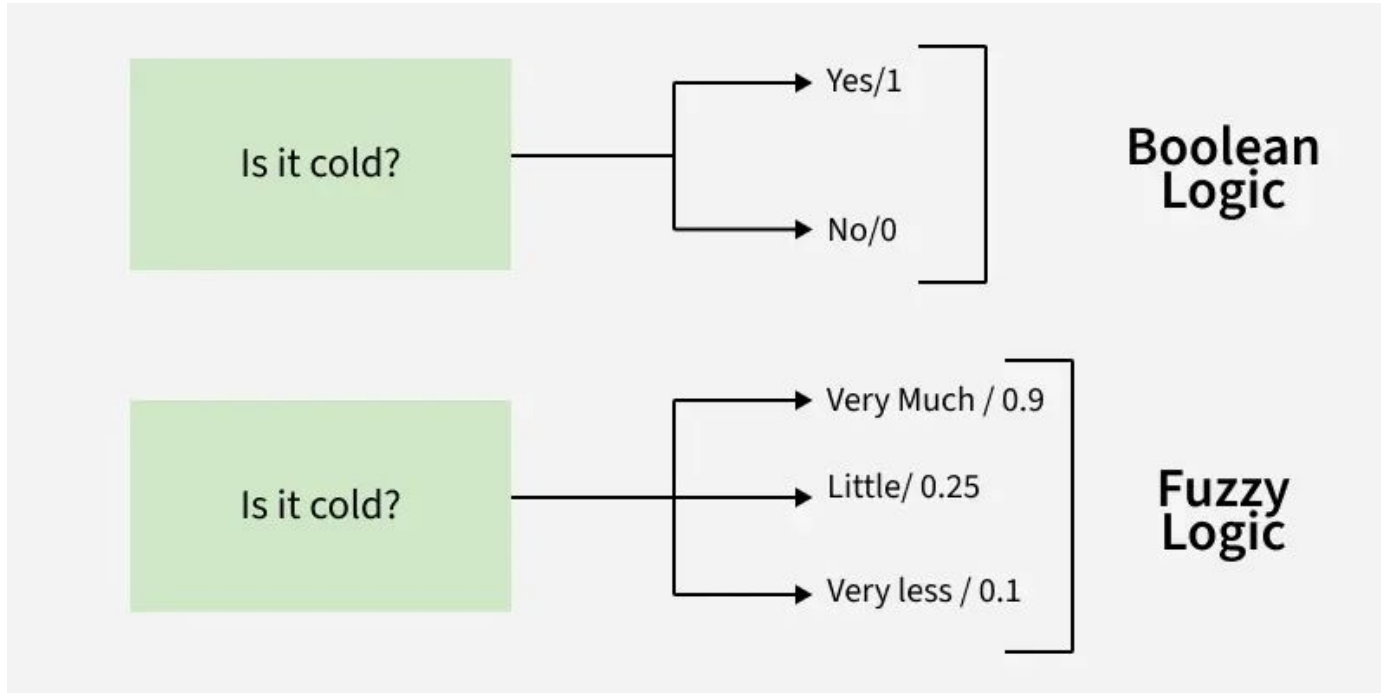
$$P(M|C) = \frac{0.04845}{0.05678}$$

$$P(M|C) \approx 0.8533$$

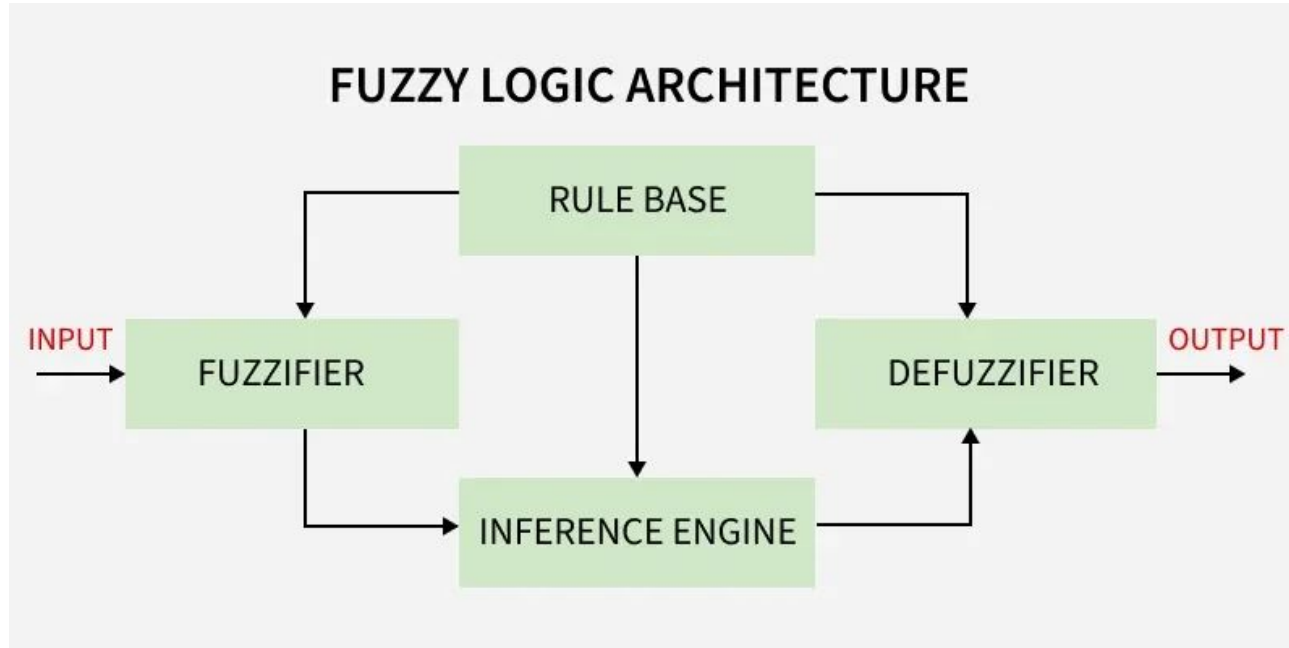
Fuzzy Logic

- Fuzzy : difficult to perceive; indistinct or vague.
- Fuzzy Logic is a form of many-valued logic in which the truth values of variables may be any real number between 0 and 1, instead of just the traditional values of true or false.
- It is used to deal with imprecise or uncertain information and is a mathematical method for representing vagueness and uncertainty in decision-making.
- It allows for partial truths, where a statement can be partially true or false, rather than fully true or false.

Fuzzy Logic



Architecture / Components of a Fuzzy Inference System



Architecture

- ❖ RULE BASE:
 - It contains the set of rules and the IF-THEN conditions provided by the experts to govern the decision-making system, on the basis of linguistic information.
- ❖ FUZZIFICATION:
 - It is used to convert inputs i.e. crisp numbers into fuzzy sets.
 - Crisp inputs are basically the exact inputs measured by sensors and passed into the control system for processing, such as temperature pressure, rpm's, etc.
- ❖ INFERENCE ENGINE:
 - It determines the matching degree of the current fuzzy input with respect to each rule and decides which rules are to be fired according to the input field.
 - Next, the fired rules are combined to form the control actions.
- ❖ DEFUZZIFICATION:
 - It is used to convert the fuzzy sets obtained by the inference engine into a crisp value.

S.No	Crisp Set	Fuzzy Set
1	Crisp set defines the value is either 0 or 1.	Fuzzy set defines the value between 0 and 1 including both 0 and 1.
2	It is also called a classical set.	It specifies the degree to which something is true.
3	It shows full membership	It shows partial membership.
4	Eg1. She is 18 years old. Eg2. Rahul is 1.6m tall	Eg1. She is about 18 years old. Eg2. Rahul is about 1.6m tall.
5	Crisp set application used for digital design.	Fuzzy set used in the fuzzy controller.
6	It is bi-valued function logic.	It is infinite valued function logic
7	Full membership means totally true/false, yes/no, 0/1.	Partial membership means true to false, yes to no, 0 to 1.

The Role of Membership Functions

A computer does not understand English words like "Tall," "Hot," or "Fast." It only understands numbers.

- The Membership Function (μ) translates a Crisp Input (e.g., Temperature = 35°C) into a Degree of Membership (e.g., 0.8 "Hot").
- It maps the input space (X-axis) to a value between 0 and 1 (Y-axis).

Common Shapes of MFs:

- Triangular MF: The simplest and most used. Good for basic control systems.
- Trapezoidal MF: Flat on top. Good when a value stays "High" for a range (e.g., "Hot" is anything between 30°C and 40°C).
- Gaussian (Bell) MF: Smooth curve. Best for natural/biological data (no sharp corners).

Fuzzy Logic System

Inputs: Temperature = 30°C and Humidity = 65%.

Steps:

- Convert crisp inputs (30°C, 65%) into fuzzy sets using the membership functions.
- Apply rules to compute fuzzy output sets.
- Aggregate fuzzy outputs and convert to a crisp value (defuzzification).

Output: Fan Speed 80%

Applications

Fuzzy Logic is used in a wide range of applications, such as **control systems, image processing, natural language processing, medical diagnosis, and artificial intelligence**